



UNIVERSITY OF THE FREE STATE

BLOEMFONTEIN CAMPUS

CSIS3744
Memorandum

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATICS

CONTACT NUMBER: 051 401 3702

EXAMINATION: Special Examination 2021

ASSESSOR: 1. Prof Okuthe P. Kogeda

MODERATOR: 1. Mnr. W.S.J. Marais

TIME: 3 HOURS

MARKS: 100

INSTRUCTIONS: Answer ALL the questions. Write legibly. Number your questions clearly and correctly.

Question 1 [29 Marks]

Vox ISP recently installed a new DNS server, dns.voxisp.co.za, on their network with the IP address, 192.168.0.254.

(a) Vox ISP wants to divide their network into 2 subnets.

i. What is the default subnet mask for Vox ISP's network? [1 Mark]

<<< 255.255.255.0 ✓>>>

- ii. Calculate the new subnet mask for the proposed sub networks. [3 Marks]

<<< 192.168.0.0 is a class C network address.

Default subnet mask for class C network is 255.255.255.0.

Binary value of subnet mask is:

11111111.11111111. 11111111.00000000

$$2^n - 2 = 2 \quad n = 2 \quad \checkmark$$

Change 2 bits from 0 to 1 for the last octet.

11111111.11111111.11111111.11000000

$$\text{Value of } 11000000 \rightarrow 2^7 + 2^6 = 128 + 64 = 192 \quad \checkmark$$

Thus new subnet mask is 255.255.255.192 $\checkmark$

OR

Default subnet mask for class C network is 255.255.255.0

Binary value of subnet mask is 11111111.11111111.11111111.00000000

$$2^n = 2 \quad n=1 \quad \checkmark$$

Change 1 bit from 0 to 1 for the last octet.

11111111.11111111.11111111.10000000 $\checkmark$

$$\text{Value of } 10000000 \Rightarrow 2^7 = 128$$

The new subnet mask is 255.255.255.128 $\checkmark$

- iii. How many usable addresses are available in the whole network? [3 Marks]

$$<<< 2^6 - 2 \checkmark = 64 - 2 = 62 \times 2 \checkmark = 124 \quad \checkmark$$

OR

$$(2^7 - 2) \checkmark = 126 \times 2 \checkmark = 252 \quad \checkmark$$

- iv. Write the above mentioned network server's IP address in CIDR notation.
[2 Marks]

$$<<< 192.168.0.254 / 25 \checkmark \checkmark \text{ OR } 192.168.0.254 / 26 >>>$$

- (b) Complete the table below with each subnet's network address, broadcast address and usable host address range: [4 Marks]

	Subnet	Network address	Broadcast address	Usable host address range
(i)	1	192.168.0.64½	192.168.0.127½	192.168.0.65½ to 192.168.0.126½
(ii)	2	192.168.0.128½	192.168.0.191½	192.168.0.129½ to 192.168.0.190½

OR

	Subnet	Network address	Broadcast address	Usable host address range
(i)	1	192.168.0.0 ½	192.168.0.127 ½	192.168.0.1 ½ to 192.168.0.126 ½
(ii)	2	192.168.0.128 ½	192.168.0.255 ½	192.168.0.129 ½ to 192.168.0.254 ½

(c) What protocol can Vox ISP use to automate the process of IP addressing? [1 Mark]

<<< DHCP ✓ >>>

(d) Explain in detail how the protocol in 1(c) operates. [3 Marks]

<<< The client sends out a DHCP discover packet in broadcast fashion via the UDP protocol to the DHCP server. ✓

Every DHCP server on the same subnet as the client receives the broadcast request and responds with an available IP address. ✓

The client accepts the first IP address that it receives, responding with a broadcast message that essentially confirms to the DHCP server that it wants to accept the address. ✓

When the selected DHCP server receives the confirmation, it replies to the client with an acknowledgment message and leases the IP address for a specific period. It also provides more information, such as DNS, subnet mask, or gateway addresses that the client might have requested. ✓>>>

(e) What console command can be used to determine whether the DNS server is responding? Write down the full command. [2 Marks]

<<< "Ping ✓ dns.voxisp.co.za ✓" OR "Ping ✓ 192.168.0.254 ✓".>>>

(f) Discuss the FOUR advantages of virtualization.

[4 Marks]

- <<< 1. Efficient use of resources. ✓
2. Cost and energy savings. ✓
3. Fault and threat isolation. ✓
4. Simple backups, recovery, and replication. ✓>>>

(g) Briefly explain the difference between a bridged, NAT and host-only network connection type with regard to virtualization.

[3 Marks]

<<< Bridged: The vNIC accesses the physical network using the host machine's NIC and obtains its own IP address, default gateway, and subnet mask from the DHCP server on physical LAN. ✓

NAT: The virtual network is hidden from the physical network but can still communicate with it. It obtains its IP address information from the virtual DHCP server on the host machine. ✓

Host-only: VMs on one host can exchange data with each other and the host but cannot communicate with nodes beyond the host. ✓>>>

(h) Discuss why network segmentation is a good idea when managing a large network.

[3 Marks]

<<< For a network administration, segmenting a network would help ease network traffic ✓ and ensures that high bandwidth is made available at all times for all users. ✓ This translates to better performance especially for a growing network. ✓>>>

Question 2 [20 Marks]

(a) What are the differences between the POP3 and IMAP4 protocols? [2 Marks]

<<< Using the POP3 (Post Office Protocol, version 3), email is downloaded to the client computer. ✓ When using IMAP4 however, the client application manages the email while it's stored on the server. ✓>>>

(b) What is the difference between WAN and MAN? [2 Marks]

<<< A group of LANs that spread over a wide geographical area is called a WAN (wide area network). ✓ A group of connected LANs in the same geographical area-for example, a handful of government offices surrounding a state capitol building-is known as a MAN (metropolitan area network) ✓>>>

(c) Explain FOUR tasks for which network operating system is responsible. [4 Marks]

<<< Managing data and other resources for a number of clients ✓
* Ensuring that only authorized users access the network ✓
* Controlling which types of files a user can open and read ✓
* Restricting when and from where users can access the network ✓
* Dictating which rules computers will use to communicate ✓
* In some situations, supplying applications and data files to clients. ✓ >>>

(d) Explain the differences between a physical topology and a logical topology. [2 Marks]

<<< The term physical topology, or network topology, mostly applies to hardware and describes how computers, other devices, and cables fit together to form the physical network. ✓ The term logical topology has to do with software and describes how access to the network is controlled, including how users and programs initially gain access to the network and how specific resources, such as applications and databases, are shared on the network. ✓ >>>

(e) Describe how the Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) differ from each other, and provide examples of where each might be used. [4 Marks]

<<< The Transmission Control Protocol (TCP) makes a connection with the end host, checks whether the data is received, and resends it if it is not. TCP is, therefore, called a connection-oriented protocol. ✓ TCP is used by applications such as Web browsers and email. Guaranteed delivery takes longer and is used when it is important to know that the data reached its destination. ✓ The User Datagram Protocol (UDP) does not guarantee delivery by first connecting and checking whether data is received; thus, UDP is called a connectionless protocol or best-effort protocol. ✓ UDP is used for broadcasting, such as streaming video or audio over the Web, where guaranteed delivery is not as important as fast transmission. UDP is also used to monitor network traffic. ✓ >>>

(f) What is electromagnetic interference, and how do you avoid its effects? [2 Marks]

<<< Electromagnetic interference is interference that is caused by motors, power lines, televisions, copiers, fluorescent lights, or other sources of electrical activity. ✓ In order to reduce the possibility of noise that can affect your network's signals, you should install cable at least 3 feet away from any EMI sources. ✓ >>>

(g) Explain the three different types of cabling recognized by TIA/EIA as acceptable for horizontal wiring. [3 Marks]

<<< TIA/EIA recognizes three possible cabling types for horizontal wiring: UTP, STP, or fiber optic cable. UTP (unshielded twisted pair) cable is a type of copper-based cable that consists of one or more insulated twisted-wire pairs encased in a plastic sheath. ✓ STP (shielded twisted pair) cable is a type of copper-based cable containing twisted-wire pairs that are not only individually insulated, but also surrounded by a shielding made of a metallic substance such as foil. ✓ Fiber-optic cable is a form of cable that contains one or several glass or plastic fibers in its core and comes in two types: single-mode fiber (SMF) or multimode fiber (MMF). Copper-based cable transmits data via electric signals, and

fiber-optic cable transmits data via pulsing light sent from a laser or light-emitting diode (LED). ✓ >>>

- (h) Explain the difference between unicast, multicast, and anycast addresses. [3 Marks]
<<< A unicast address specifies a single node on a network. ✓ This differs from a multicast address, which is delivered to all nodes in a targeted, multicast group. ✓ An anycast address can identify multiple destinations, with packets delivered to the closest destination. ✓ For example, a DNS name server might send a DNS request to a group of DNS servers that have all been assigned the same anycast address. A router handling the request examines routes to all the DNS servers in the group and routes the request to the closest server. >>>

Question 3 [27 Marks]

Network addressing is very important.

- (a) Discuss the two different variations of network address translation (NAT). [4 Marks]

<<< The two variations of NAT are:

* SNAT-Using SNAT (Static Network Address Translation or Source Network Address Translation), the gateway assigns the same public IP address to a host each time it makes a request to access the Internet. Small home networks with only a single public IP address provided by its ISP use SNAT. ✓✓

* DNAT or Destination NAT-Using DNAT (Destination Network Address Translation), hosts outside the network address a computer inside the network by a predefined public IP address. When a message sent to the public IP address reaches the router managing DNAT, the destination IP address is changed to the private IP address of the host inside the network. The router must maintain a translation table of public IP addresses mapped to various hosts inside the network. ✓✓ >>>

- (b) Explain the difference between a port and socket. [4 Marks]

<<< A port is a number assigned to a process, such as an application or a service, that can receive data. Whereas an IP address is used to find a computer, a port is used to find a process running on that computer. TCP and UDP ports ensure that data is transmitted to the correct process among multiple processes running on the computer. ✓✓ A socket consists of both a host's IP address and a process's TCP or UDP port, with a colon separating the two values. For example, the standard port for the Telnet service is TCP 23. If a host has an IP address of 10.43.3.87, the socket address for Telnet running on that host is 10.43.3.87:23. ✓✓>>>

- (c) Discuss the two different types of DNS requests. [4 Marks]

<<< There are two types of DNS requests:

- * recursive query-A query that demands a resolution or the answer "It can't be found." For example, the initial request the resolver makes to the local server is a recursive query. The local server must provide the information requested by the resolver, as in "The buck stops here." ✓✓
- * iterative query-A query that does not demand resolution. For example, when the local server issues queries to other servers, the other servers only provide information if they have it. ✓✓ >>>

(d) Explain some examples of routing metrics that can be used to determine the best path for a network. [4 Marks]

<<< Some examples of routing metrics used to determine the best path may include:

- * Hop count, which is the number of network segments crossed ✓
- * Theoretical bandwidth and actual throughput on a potential path ✓
- * Delay, or latency, on a potential path, which results in slower performance ✓
- * Load, which is the traffic or processing burden sustained by a router in the path ✓
- * MTU, which is the largest IP packet size in bytes allowed by routers in the path without fragmentation (excludes the frame size on the local network) ✓
- * Routing cost, which is a value assigned to a particular route as judged by the network administrator; the more desirable the path, the lower its cost ✓
- * Reliability of a potential path, based on historical performance ✓
- * A network's topology. ✓ >>>

(e) Discuss FOUR basic functions of all network router. [4 Marks]

<<< Although any one router can be specialized for a variety of tasks, all routers can do the following:

- * Connect dissimilar networks, such as a LAN and a WAN, which use different types of routing protocols. ✓
- * Interpret Layer 3 and often Layer 4 addressing and other information (such as quality of service indicators). ✓
- * Determine the best path for data to follow from point A to point B. The best path is the most efficient route to the message's destination calculated by the router, based upon the information the router has available to it. ✓
- * Reroute traffic if the path of first choice is down but another path is available. ✓ >>>

(f) Noise can degrade or distort a signal on a network. Discuss the two common causes of noise. [4 Marks]

<<< Two common sources of noise are:

- * EMI (electromagnetic interference)-Caused by motors, power lines, televisions, copiers, fluorescent lights, microwave ovens, manufacturing machinery, and other sources of electrical activity (including a severe thunderstorm). One type of EMI is RFI (radio frequency interference), or electromagnetic interference caused by radio waves. (Often, you'll see EMI referred to as EMI/RFI.) Strong broadcast signals from radio or

TV antennas can generate RFI. ✓✓

* crosstalk-Occurs when a signal traveling on one wire or cable infringes on the signal traveling over an adjacent wire or cable. The resulting noise, or crosstalk, is equal to a portion of the second line's signal. If you've ever been on a traditional, landline phone and heard the conversation on a second line in the background, you have heard the effects of crosstalk. ✓✓ >>>

- (g) When deciding between omnidirectional and unidirectional antenna types, why might the use of an omnidirectional antenna be inadvisable? [3 Marks]

<<< You might think that omnidirectional antennas would nearly always be the best choice when setting up Wi-Fi coverage. The idea is to place the AP in the center of its coverage area, then send the signal out in all directions. However, in many situations, installing unidirectional antennas instead will enhance a signal's availability✓, directing the signal right where you need it while not wasting a signal in areas where you don't. ✓ For example, suppose a company installs an omnidirectional antenna near a factory's 30-foot high ceiling. Because the antenna's signal is broadcast in all directions from its location, distributing its signal strength in a spherical shape, the best possible signal would only be available to workers who could walk on the ceiling-obviously, that's not a viable situation. To be useful, the signal needs to be directed down to the floor. A unidirectional antenna, in this case, can be positioned up high and pointed down to create a dome-shaped coverage that spreads out as it nears the plant floor. ✓ >>>

Question 4 [22 Marks]

There are three primary areas or classifications of security controls. These include management security, operational security, and physical security controls.

- (a) Explain the difference between war driving and war chalking. [2 Marks]

<<< In war driving, a hacker searches for unprotected wireless networks by driving around with a laptop or smartphone configured to receive and capture wireless data transmissions. (The term is derived from the term war dialing, which is a similar tactic involving old, dial-up modems.) War driving is surprisingly effective for obtaining private information. ✓ War chalking occurs when a hacker discovers vulnerable access points, and makes this information public by drawing symbols with chalk on the sidewalk or wall within range of a wireless network. The symbols, patterned after marks that hobos devised to indicate hospitable places for food or rest, indicate the access point's SSID and whether it's secured. Alternatively, many websites offer maps of these open networks, as reported by war drivers. ✓ >>>

(b) Most wireless devices use either FHSS or DSSS technology to avoid interference.

Explain these two technologies.

[4

Marks]

<<< The two technologies used are as follows:

* FHSS (frequency hopping spread spectrum)-Short bursts of data are transmitted on a particular frequency within the band and the next burst goes to the next frequency in the sequence. Frequency hopping can happen hundreds of times a second. FHSS is cheaper to implement than DSSS and performs better than DSSS in crowded, indoor environments.

✓✓

* DSSS (direct sequence spread spectrum)-Data streams are divided and encoded into small chunks, called chips, which are spread over all available frequencies within one of three, wide channels, all at the same time. The process of dividing and encoding the data is called chipping, and the spreading ratio used to transform the data is called the chipping code, which is unique to each device. DSSS uses the available bandwidth more efficiently than FHSS and tends to have a higher throughput. ✓✓ >>>

(c) Discuss FOUR characteristics of malware that make it difficult to detect. [4

Marks]

<<< Some of the characteristics that make malware harder to detect and eliminate include the following:

* encryption-Some malware is encrypted to prevent detection. Most antimalware software searches files for a recognizable string of characters that identify the virus. However, encryption can thwart the anti-malware program's attempts to detect it. ✓

* stealth-Some malware disguises itself as legitimate programs or replaces part of a legitimate program's code with destructive code. ✓

* polymorphism-Polymorphic malware changes its characteristics (such as the arrangement of bytes, size, and internal instructions) every time it's transferred to a new system, making it harder to identify. ✓

* time dependence-Some malware is programmed to activate on a particular date. This type of malware can remain dormant and harmless until its activation date arrives. Time-dependent malware can include logic bombs, or programs designed to start when certain

conditions are met. (Logic bombs can also activate when other types of conditions, such as a specific change to a file, are met, and they are not always malicious.) ✓>>>

- (d) Discuss the THREE components required to manage access control to a network and its resources. [3

Marks]

<<< The three components required are as follows:

* authentication-Authentication, in this case, user authentication, is the process of verifying a user's credentials (typically a username and password) to grant the user access to secured resources on a system or network. In other words, authentication asks the question, "Who are you?" ✓

* authorization-Once a user has access to the network, the authorization process determines what the user can and cannot do with network resources. In other words, authorization asks the question, "What are you allowed to do?" Authorization restrictions affect Layer 2 segmentation, Layer 3 filtering, and Layer 7 entitlements. For example, what VLAN are you assigned to? What servers or databases can you access? What commands can you run on a device? ✓

* accounting-The accounting system logs users' access and activities on the network. In other words, accounting asks, "What did you do?" The records that are kept in these logs are later audited, either internally or by an outside entity, to ensure compliance with existing organizational rules or external laws and requirements. ✓>>>

- (e) In preparing and planning for disaster recovery, explain the difference between an incident and a disaster. [2

Marks]

<<< An incident is any event, large or small, that has adverse effects on a network's availability or resources. This could be a security breach, such as a hacker gaining access to a user's account, an infection, such as a worm or virus, or an environmental issue, such as a fire or flood. ✓ A disaster on the other hand is an extreme type of incident, involving a network outage that affects more than a single system or limited group of users. ✓>>>

- (f) Discuss FOUR common network performance metrics that are utilized. [4 Marks]

<<< Some of the more common performance metrics are as follows:

- * utilization-This metric refers to the actual throughput used as a percentage of available bandwidth. No network should operate at maximum capacity. Identify patterns of utilization and ensure that available bandwidth accounts for utilization spikes. ✓
- * error rate-Bits can be damaged in transit due to EMI or other interference. The calculated percentage of how often this occurs is the error rate. ✓
- * packet drops-Packets that are damaged beyond use, arrive after their expiration, or are not allowed through an interface are dropped. Packet drops result in delayed network communications while devices wait for responses or have to resend transmissions. Knowing what's normal for your network will help you identify problems when packet drop rates vary. ✓
- * jitter-All packets experience some latency. When successive packets experience varying amounts of latency, resulting in their arriving out of order, the user experience is degraded. This is called jitter, a problem that can be addressed through traffic management techniques. ✓ >>>

- (g) Discuss how ATM is able to guarantee a specific QoS for certain types of transmissions. [3 Marks]

<<< Establishing a reliable connection allows ATM to guarantee a specific QoS for certain types of transmissions. ATM networks can supply four QoS levels, from a "best effort" attempt for noncritical data to a guaranteed, real-time transmission for time sensitive data. ✓ This is important for organizations using networks for time-sensitive applications, such as video and audio transmissions. ✓ For example, a company might want to use its ATM connection between two offices located at opposite sides of a state to carry voice phone calls with the highest possible QoS. On the other hand, the company might assign a low QoS to routine email messages exchanged between the two offices. ✓ >>>

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